SRIKANTH MULUPOJU

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SUMMARY

Senior Hardware Design Engineer with 12 years of experience in automotive R&D specializing in hardware design, high-speed PCB development, microcontroller-based systems, power electronics, and EMI/EMC compliance. Proven expertise in product feasibility analysis, DFMEA, board bring-up, debugging, and cross-functional collaboration across global teams. Experienced in automotive standards (ISO, CISPR, IATF 16949, ASPICE) and delivering cost-optimized, reliable hardware solutions for ECUs, media hubs, LED lighting systems, wireless chargers, Audio amplifier and DC-DC converters.

SKILLS

Architecture & Leadership: Hardware Architecture Definition, RFQ Technical Review, Feasibility Analysis, Cost Estimation, Requirement Analysis, System Partitioning, Risk Assessment, Customer Technical Discussions, Technical Mentoring
Hardware Design & Development: High-Speed PCB Design (4+ Layers), Microprocessor/Microcontroller Design, Power Supply Design, DC-DC Converters, Automotive ECUs, Sensor Interface, Automotive Media Hubs
Protocols & Interfaces: CAN, LIN, SPI, I2C, USB; exposure to HDMI-based systems
Simulation & Analysis: LTSpice, MATLAB, Thermal Calculations, Power Dissipation Analysis, EMI/EMC Analysis
EMI/EMC & Standards: ISO, CISPR, JASO, IEC, ASPICE, IATF 16949, DFMEA, Reliability Testing
EDA & Tools: Altium Designer, OrCAD Capture/PCB Designer, Fusion 360, Ansys (SI wave, HFSS, Q3D, AEDT), CANalyzer, Vector CANoe, GRL Automation Software for Qi testing
Debugging & Validation: Oscilloscope, Logic Analyzer, Board Bring-up, Manufacturing Support, Root Cause Analysis
Programming: C, Python, MATLAB
Data Management: Part Analytics, BOM Lifecycle Management, Plex Manufacturing ERP
Configuration Control: Git/SVN-based Version Control, Revision Management
Soft Skills: Technical Leadership, Cross-functional Collaboration, Supplier Coordination, Global Stakeholder Communication

TECHNICAL LEADERSHIP & ARCHITECTURE EXPERIENCE

- Led hardware architecture definition for automotive electronics products from RFQ stage to production.
- Conducted requirement analysis and converted customer specifications into block diagrams and system-level schematics.
- Experience in automotive electrical transient protection and reliability-driven design.
- Performed risk analysis (DFMEA) and reliability planning during early design phase.
- Coordinated with cross-functional teams
- Provided technical guidance and mentoring to junior engineers.
- Conducted benchmark studies on competitor automotive electronic products
- Performed Product teardown analysis and component comparison
- Evaluated circuit architecture, component selection, and cost structure
- Generated cost optimization strategies through component comparison and redesign proposals

PROFESSIONAL EXPERIENCE

AMPHENOL | Hyderabad *(Mar 2021 – Present)*

Sr. R&D Engineer

- Supported hardware solutions for major 4-wheeler OEM programs
- Participated in **RFQ meetings** with customers and internal stakeholders to analyze technical requirements and propose optimized hardware solutions.
- Performed schematic design, footprint creation, PCB layout review, and board bring-up.
- Conducted DFMEA, reliability validation, and EMI/EMC compliance testing.
- Supported prototype builds, debugging, rework, and manufacturing issue resolution.
- Used Altium 365 for cloud-based schematic/PCB collaboration and release management
- Managed component lifecycle and obsolescence risk using Part Analytics
- Maintained schematic & PCB revisions using structured version control methodology
- Ensured design traceability and document control as per automotive quality standards
- Supported controlled design releases for OEM programs

Projects:

- Developed automotive-grade **Qi2.0 wireless charger** systems ensuring thermal stability and EMI compliance.
- Designed USB Type-C & Power Delivery (PD) fast charging solutions for automotive infotainment systems.
- Led design and development of Automotive Media Hub integrating USB, HDMI, audio, and SSD interfaces.
- Automotive Auxiliary Control Panel (Multi-Gang Toggle Switch System)

R&D Engineer

- Designed schematic and PCB layout using Autodesk EAGLE
- Performed 3D PCB integration and enclosure validation using Fusion 360
- Conducted mechanical clearance checks for housing and connector fitment
- Verified mounting points, standoff alignment, and board-to-enclosure tolerances
- Coordinated ECAD–MCAD integration for optimized thermal and space utilization
- Supported prototype mechanical inspection of PCBA assemblies
- Designed low and ultra-low pressure sensors with high accuracy and repeatability for automotive applications.

MAGNETI MARELLI MOTHERSON AUTOMOTIVE | Pune *(Aug 2019 – Mar 2021)*

Hardware Design Engineer

- Designed DC-DC converters and automotive power supply systems.
- Designed load dump & ISO 7637 compliant protection circuitry
- Implemented 4-layer PCB designs using OrCAD Capture and PCB Designer.
- Conducted ECU testing and diagnostics using Vector CANoe/CANalyzer tools.
- Performed power calculations, schematic validation, and thermal estimations.
- Coordinated with mechanical teams for PCB area definition and enclosure integration.
- Supported BOM preparation, supplier technical discussions, and customer queries.
- Provided inputs for VAVE activities and alternate component selection
- Identified design optimization opportunities for performance and cost improvement
- Conducted competitive benchmarking of LED headlamp assemblies from major OEM vehicles

Projects:

- LED Headlamp ECU
- Rear Combination Lamps

STAR ENGINEERS INDIA PVT. LTD | Pune *(Jan 2014 – Aug 2019)*

R&D Engineer – Automotive Electronics

- Designed Analog and microcontroller-based automotive products.
- Applied EMI/EMC standards (ISO, CISPR, JASO, IEC) for product validation.
- Worked closely with OEMs including Honda, Bajaj, Suzuki, Piaggio, and Vinfast.
- Supported emission lab testing and regulatory certification.
- Collaborated with testing teams to meet functional, thermal, and mechanical specifications.
- Coordinated ARAI compliance visits for automotive product validation and certification
- Supported EMI/EMC and regulatory compliance testing activities
- Performed VAVE (Value Analysis & Value Engineering) to optimize BOM cost and improve design efficiency
- Identified and qualified alternate components to mitigate supply chain risks
- Handled multiple automotive projects from RFQ to production release

Key Projects:

- Capacitor Discharge Ignition (CDI) Systems
- Headlamp Control Unit for 2-wheelers
- Blinker with Hazard Control System
- Side stand safety switch circuit for two-wheeler application
- Ignition start/stop switch control circuit for automotive application

KEY COMPETENCIES – AUTOMOTIVE ELECTRONICS & HARDWARE ARCHITECTURE

- Hardware feasibility analysis for Automotive Embedded systems
- High-speed PCB routing and signal integrity awareness
- Microcontroller and automotive communication design
- EMI/EMC problem solving as per automotive standards
- DFMEA and reliability-driven product design
- Prototype validation, debugging & root cause analysis
- Cross-functional collaboration (Mechanical, Software, Architecture Teams)
- Technical mentoring and team coordination

EDUCATION

- MBA, Manipur International University
- Executive Development Program in Project Management – XLRI
- B.Tech in Electronics & Communication Engineering – JNTU Hyderabad

CERTIFICATIONS & TRAINING

- IPC Certified Interconnect Designer (CID)
- ESD Certification – Electronics Systems & Devices
- Training on TS16949 & IATF 16949
- Embedded Systems Certification
- ASPICE Assessment Training
- Automotive Functional Safety awareness (ISO 26262)

ADDITIONAL ACTIVITIES

- Volunteer Member – Brahma Kumaris
  Participate in meditation and self-development programs focused on discipline and ethical practices